

Value [SimTalk] - Checkbox

Activates (*true*) or deactivates (*false*) the *Checkbox* designated by <Path>. The *Checkbox* executes a **Control** when its value changes.

Syntax

```
<Path>.Value:boolean
```

Watchable

The attribute is **watchable**.

Assignment Value

You can assign a value of data type *boolean*.

Example

```
MyCheckbox.Value := false
```

See also

[Value \[drop-down list\] - Checkbox](#)

[Control \[Checkbox\]](#)

Button

Button [object]

Use the object *Button* for showing a button in the *Frame*. When you click the button, it executes the action which you programmed in a control.

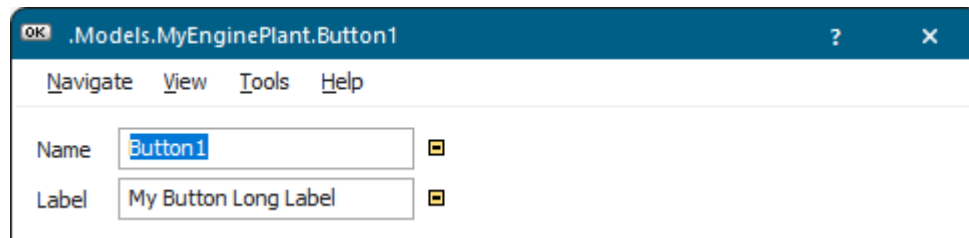
Image



Description

The *Button* executes the action, which you programmed in the **Control**, when the user clicks it.

If you type in a label for the button, *Plant Simulation* shows that **label** on the button in the *Frame*.



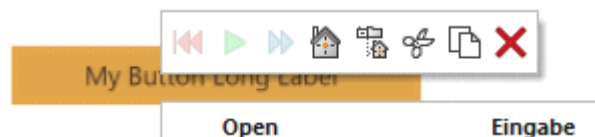
By default, *Plant Simulation* does not show the name of the button as a *tooltip* in the *Frame* when you roll the mouse over it. To show your own tooltip, create a *user-defined attribute* and name it `ToolTip`.



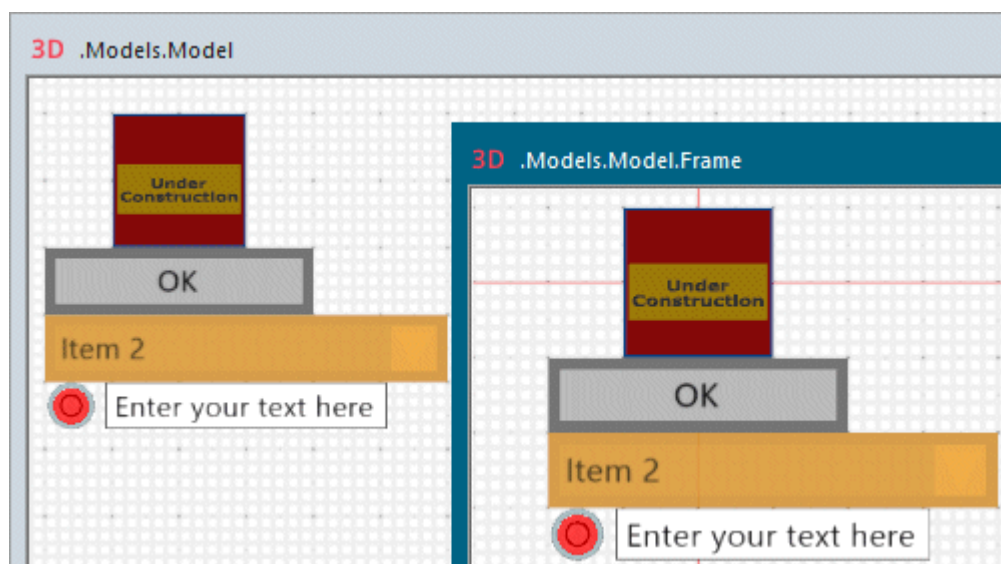
Plant Simulation can show the button in the *Frame* in a number of ways:

- With the label you type in .
- With an **icon**  you draw in the *icon editor*. Then, you can show text you type in as the label on that button,  for example.
- With a **picture** you pasted into a new icon in the *icon editor*,  and  for example. You could name the icons `play` and `play_down` for example.

To open its dialog box, click the button with the right mouse button and select **Open** on the context menu.



You can also click a *Button* in a *sub-Frame* and trigger the respective action.



To select the button in the *Frame* window, you can also drag a marquee over its icon.

By default, the *Button* looks like this:



Note

Plant Simulation only accepts the click the *Button* if it occupies at least 10 pixels in each direction in the model window. If you zoom out very far, the picture of the *Button* might become too small, preventing *Plant Simulation* to recognize it reliably.

To show a **tooltip** with information about the *Button*, hover with the mouse over it.

To change the length of the graphic and the anchor points of the *Button*, click **Show Manipulators** on the **Edit** ribbon tab or press **M** on the keyboard.



Add the Object to the Simulation Model

To add the object *Button* to your simulation model, click  [Manage Class Library](#) > **Basic Objects** > **UserInterface** > **Button** on the **Home** ribbon tab.

See also

[Execute the Action](#)

[Select an Option from a Drop-down List](#)

[Configure the Buttons to Open Parts Table and Callback Method](#)

Dialog Box of the Button

Dialog Box of the Button

Click the *Button*, which you inserted into a *Frame*, with the right mouse button and select **Open** on the context menu to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under [Dialog Items of the Objects](#).

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box [Edit 3D Properties](#):

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click [Show Manipulators](#) on the **Edit** ribbon tab or press **M** on the keyboard.

Tab Attributes

Tab Attributes

The tab **Attributes** provides the settings, which the object offers.

The settings are listed in the table of contents to the left.

The shared properties are described under the [Tab Attributes](#).

Width [text box] - Button

Type in the **Width** in meters with which the object is displayed in the *Frame*. If you type in long text, you have to adjust the width so that the text fits into the object boundaries without being cut off.

Width

■ m

My Button Long Label

Remarks

Plant Simulation shows the **Width** in meters.

SimTalk

ObjectWidth [SimTalk] - Button

Height [text box] - Button

Type in the **Height** in meters with which the object is displayed in the *Frame*. To change the height, you can also hold down **Ctrl+Shift** and drag the top or the bottom of the icon.

Height

m

My Button Long Label

Remarks

To change the height, you can also hold down **Ctrl+Shift** and drag the top or the bottom of the icon.

Plant Simulation shows the **Height** in meters.

SimTalk

ObjectHeight [SimTalk] - Button

Control [Button]

Modifies the built-in behavior of the object. The object calls the **Control** when you click the *Button*.



Remarks

The anonymous identifiers ? and @ are set to the object when the method is called. Specify a control without parameter to call the control one time when you release the *Button*.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

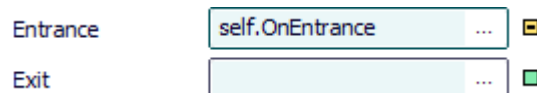
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select [Create Control \[context menu\]](#). *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

Parameter

You can also specify a control which expects a *boolean* value as a parameter. This control is called with the value `true` when you click the *Button* and with the value `false` when you release it.

Plant Simulation executes the **Control** when you change the size of the *Button* or when you move it with Drag-and-Drop. If the control does not expect a parameter it will be called, as before, when you release the *Button*.

Example

A control might, for example, look like this:

```
self.~.~.DataTable.openDialog
```

```
self.~.~.&MyMethod.openDialog // opens the dialog of the
// method MyMethod, instead of executing its source code
end
```

SimTalk

Control [SimTalk] - Button

See also

[Select Object \[for controls\]](#)

Tab User-defined

Define your own attributes as described under the [Tab User-defined](#).

By default *Plant Simulation* does not show the name of the button as a tooltip in the *Frame* window. To show a your own tooltip, you can create a user-defined attribute and name it `Tooltip`.



Navigate Menu

The commands are described under the [Navigate Menu](#).

View Menu

The commands are described under the [View Menu](#).

SimTalk

[updateDialog \[SimTalk\]](#)

Tools Menu

The **Tools Menu** provides commands to access its functions.

[Edit Controls](#)

[Edit Observers](#)

See also

[Tools Menu \[general description\]](#)

Help Menu

The commands are described under the [Help Menu](#).

Methods of the Button

Methods of the Button

The *Button* provides the [Methods of All Objects](#).

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window [Show Attributes and Methods](#). The figure below illustrates the information using the example of the object *Station*.

Name	Value	Inherited	Watchable	Signature
contentsList	[]			((Contents:table)) -> any
createAttr				(NameOfUserDefinedAttribute:string, DataType:string) -> boolean
createIcon				(IconName:string, Width:integer, Height:integer) -> integer
deleteAttr				(NameOfUserDefinedAttribute:string) -> boolean

Name of the method Identifier and data type of the parameter you enter Data type of the return value

Name	Value	Inherited	Watchable	Signature
moveToFolder				(Folder:object) -> object
MU	VOID			((Index:integer:=1)) -> object
MUPart	VOID			((Index:integer:=1)) -> object

Name of the method Identifier and data type of the parameter you enter Default value Data type of the return value

- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected [Instance \[general description\]](#).

An example of the **Syntax** line of the individual methods might look like this:

```
<Path>.openDialog([CallOpenControl:boolean:=false]) → boolean
```